

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

Comparative Analysis of Network Topologies

1. Star Topology vs Bus Topology — Cost, Reliability and Performance

Star and bus topologies differ significantly in terms of cost, reliability, and performance.

Bus topology is generally less expensive because all devices share a single backbone cable and do not require a dedicated connection to a central switch. This reduces cabling and hardware costs, making it suitable for very small networks with limited budgets. However, the shared communication medium can become a bottleneck as more devices are added.

Star topology requires a central switch or hub and individual cables for every device, so the initial installation cost is higher. Despite this, star topology provides better reliability because the failure of one cable or workstation normally affects only that particular device. In a bus network, damage to the backbone can disrupt communication for all connected devices.

Performance is also generally better in a switched star network because each device has a dedicated connection to the switch, reducing contention and collisions. For example, a modern office with 20 computers would benefit more from a switched star topology than a traditional bus network.

Conclusion: Bus topology is advantageous when low initial cost is the primary concern, whereas star topology provides better reliability, performance, troubleshooting, and scalability. Therefore, star topology is generally preferred for modern LANs.

2. Mesh Topology vs Star Topology — Fault Tolerance and Reliability

Mesh topology provides greater fault tolerance than star topology because it offers multiple paths between network devices. In a **full mesh**, devices have direct connections to every other device, while a **partial mesh** provides redundant connections where they are most important.

If one link fails in a mesh network, traffic can use an alternative path. Therefore, communication can continue even during certain link failures. This makes mesh topology particularly useful in critical environments such as banking, telecommunications, data centers, and emergency services.

Star topology also provides some fault isolation. If a single workstation cable fails, other workstations can continue operating. However, the central switch represents a critical point. If that switch fails and there is no redundancy, all connected devices may lose network connectivity.

For example, a bank's core network may use redundant mesh connections to maintain connectivity during equipment or link failures, while an ordinary office may use a star topology because its availability requirements are lower.

Conclusion: Mesh topology offers superior fault tolerance and redundancy, but it is more expensive and complex. Star topology provides a better balance of reliability, cost, and simplicity for most LAN environments.

3. Bus Topology vs Mesh Topology — Installation Complexity

Bus topology is relatively simple to install because devices share a common backbone. A traditional bus network requires a backbone cable and appropriate connectors and terminators. The number of physical connections is much lower than in a full mesh network.

Mesh topology is considerably more complex. In a **full mesh**, every device requires a direct connection to every other device. The number of links required can be calculated using:

where n is the number of devices.

For example, with 5 devices:

With 10 devices:

This demonstrates how rapidly cabling and port requirements increase as the network grows.

Bus topology therefore requires less cabling and is easier to initially install, whereas mesh provides much greater redundancy at the cost of complexity.

Conclusion: Bus topology is simpler and cheaper to deploy, while mesh topology is more complex but provides superior reliability and alternative communication paths.

4. Hybrid Topology vs Star Topology — Scalability

Hybrid topology combines two or more topology types, such as star, mesh, ring, or bus, to meet different organizational requirements. This makes it highly flexible and scalable.

For example, a university campus can use a **mesh or ring structure for its high-speed backbone**, while individual buildings use **star topology** for connecting computers and access points. A new building can be added by connecting its network to the existing campus backbone without redesigning every existing building. Star topology is also relatively scalable, particularly when managed switches are used.

However, expansion is initially limited by the number of available switch ports. Additional switches can be added to overcome this limitation, but this increases hardware and configuration requirements.

Hybrid topology therefore provides greater architectural flexibility because different parts of the organization can use different topologies according to their requirements.

Conclusion: Star topology is simpler and easier to expand within a small or medium LAN, while hybrid topology provides greater scalability for large organizations, campuses, and enterprise networks.

5. Star, Mesh, Bus and Hybrid Topologies — Maintenance

Maintenance requirements vary considerably among network topologies.

Star topology is generally easiest to maintain. Because each device has an individual connection to the central switch, technicians can isolate problems to a particular cable, port, or device. For example, if one computer loses connectivity, other computers can continue operating while that connection is tested.

Mesh topology is highly reliable but more difficult to maintain because it contains many interconnected links. Troubleshooting, monitoring, and replacing components require greater technical expertise.

Bus topology has relatively simple physical construction but can be difficult to troubleshoot when the backbone develops a fault. A problem in the shared cable can affect multiple devices, making fault identification more challenging.

Hybrid topology has moderate to high maintenance complexity because different sections may use different technologies and topologies. However, its modular design can make it easier to manage large networks when proper documentation and monitoring systems are used.

Maintenance Comparison

Topology	Maintenance	Main Reason
Star	Easy	Individual links make fault isolation simple
Bus	Difficult	Backbone failure can affect the entire segment
Mesh	Very difficult	Large number of interconnected links
Hybrid	Moderate	Multiple topology types require coordinated management

Conclusion: Star topology generally provides the best balance between reliability and ease of maintenance. Mesh offers maximum redundancy but requires more maintenance effort, while bus is difficult to troubleshoot when backbone problems occur.

Overall Comparative Summary

Parameter	Star	Bus	Mesh	Hybrid
Cost	Medium	Low	Very High	Medium–High
Reliability	High	Low	Very High	High
Fault Tolerance	Moderate	Low	Very High	High
Installation	Easy	Easy	Complex	Moderate
Scalability	High	Low	Low–Moderate	Very High
Maintenance	Easy	Difficult	Difficult	Moderate
Performance	High	Low–Moderate	Very High	High
Best Use	Office LAN	Small/legacy networks	Critical networks	Large enterprises/campuses

Overall Conclusion

No single topology is ideal for every network. **Bus topology** is economical but has limited reliability and scalability. **Star topology** provides a strong balance of cost, performance, reliability, and easy maintenance, making it the most common choice for modern LANs. **Mesh topology** provides the highest redundancy and fault tolerance and is therefore suitable for critical infrastructure. **Hybrid topology** combines the strengths of multiple topologies and is best suited for large, complex, and scalable networks such as universities, enterprises, and campus environments.

Thus, topology selection should be based on **cost, reliability, scalability, performance, installation complexity, maintenance requirements, and the criticality of the network.**